

Longitudinal Development of Electromagnetic Showers in Aluminium and Lead: a Geant4 Monte Carlo Study

Florin Manea and Eric Oprea

We present a Monte Carlo study of electromagnetic shower development induced by 5 GeV electrons in aluminium and lead, performed with the Geant4 simulation toolkit. The longitudinal profiles of deposited energy and secondary-particle multiplicity are reconstructed as a function of depth expressed in radiation-length units and compared between the two absorbers. The simulated shower maxima are found to agree with the analytical predictions of the Rossi model and its empirical refinements to within $0.3 X_0$. The near-universal scaling of the longitudinal profile is confirmed, and the residual difference between the two materials is shown to follow directly from their critical energies.

I. INTRODUCTION

When a high-energy electron enters a dense medium it initiates an electromagnetic cascade through the interplay of bremsstrahlung and pair production. The resulting shower multiplies until the energy of the secondary particles falls below the critical energy E_c , at which point ionization losses and Compton scattering become dominant and the shower is absorbed. The precise characterisation of this process is central to the design of calorimeters in modern particle physics experiments.

The longitudinal development of the shower is governed by the radiation length X_0 of the absorber. A key prediction of the Rossi model is that, when depth is expressed in units of X_0 , the longitudinal profile becomes approximately universal across different materials. Since the critical energy depends on the atomic number Z , a comparison of aluminium ($Z = 13$) and lead ($Z = 82$) at the same primary energy provides a direct test of this scaling and of its material-dependent corrections.

In this work we simulate 5 GeV electrons impinging on blocks of aluminium and lead of identical geometry using the Geant4 simulation framework. The longitudinal profiles of deposited energy and secondary-particle multiplicity are reconstructed in slices of $0.1 X_0$ and compared, both with each other and with the analytical predictions of the Rossi model [1] and the empirical parameterisations adopted by the Particle Data Group [2].

II. THEORETICAL BACKGROUND

The development of an electromagnetic shower in matter is governed by two material-dependent scales: the radiation length X_0 and the critical energy E_c . The radiation length is defined as the mean distance over which an electron loses all but $1/e$ of its energy by bremsstrahlung; it also equals $7/9$ of the mean free path for pair production by a photon. An approximate expression for X_0 as a function of the atomic number Z and mass number A of the absorber is

$$X_0 = \frac{A}{4 \alpha N_A Z(Z+1) r_e^2 \ln(183 Z^{-1/3})}, \quad (1)$$

where r_e is the classical electron radius and N_A is Avogadro's number. The critical energy, at which ionization losses equal bremsstrahlung losses for electrons, is well approximated by

$$E_c = \frac{610 \text{ MeV}}{Z + 1.24}. \quad (2)$$

In the simple Rossi model, an electron with energy $E > E_c$ emits after one X_0 a photon carrying on average half its energy, and a photon with $E > E_c$ produces an e^+e^- pair after one X_0 , each particle carrying half the photon energy. The multiplication continues as long as the secondary energy exceeds E_c ; below this threshold ionization losses for electrons and Compton scattering for photons become dominant and the shower is absorbed. Within this model the total number of particles and the mean energy per particle after $t = x/X_0$ radiation lengths are

$$N(t) = 2^t, \quad E(t) = \frac{E_0}{2^t}, \quad (3)$$

so that the shower reaches its maximum, $E(t) = E_c$, at a depth

$$t_{\max} = \frac{\ln(E_0/E_c)}{\ln 2}. \quad (4)$$

A more accurate empirical parameterisation, which also distinguishes an electron from a photon-initiated showers, reads

$$t_{\max} = \ln\left(\frac{E_0}{E_c}\right) + C_{e\gamma}, \quad (5)$$

with $C_{e\gamma} = -0.5$ for an incident electron and $+0.5$ for an incident photon. The depth required to contain 95% of the shower energy is given empirically by

$$L_{95\%} = t_{\max} + 0.08 Z + 9.6 \quad [X_0 \text{ units}]. \quad (6)$$

For the two absorbers considered here, the relevant parameters and the predictions of Eq. (5) for a 5 GeV electron ($E_0 = 5000$ MeV) are summarised in Table I. The $\sim 1.8 X_0$ difference in $t_{\max}$ between the two materials is entirely accounted for by the ratio of their critical energies, $\ln(E_c^{\text{Al}}/E_c^{\text{Pb}}) = \ln(42.7/7.3) \simeq 1.77$.

Material	Z	E_c (MeV)	X_0 (cm)	$t_{\max}^{\text{pred}}$
Aluminium	13	42.7	8.9	4.26
Lead	82	7.3	0.56	6.03

TABLE I. Material parameters and analytical shower maxima [Eq. (5)] for 5 GeV electrons.

III. SIMULATION SETUP

The simulations were carried out with the Geant4 toolkit [3], starting from the `AnaEx01` example of C. Mancini Terracciano. The detector geometry, the primary beam, and the data readout were configured as follows.

The geometry was set directly in the detector-construction stage as a single homogeneous block of $250 \times 250 \times 250$ cm³. This volume is large enough to fully contain a 5 GeV shower in both materials: with $X_0^{\text{Al}} = 8.9$ cm and $X_0^{\text{Pb}} = 0.56$ cm, Eq. (6) gives 95% containment lengths of $\sim 15 X_0$ (~ 133 cm) for aluminium and $\sim 22 X_0$ (~ 12 cm) for lead, both within the simulated depth along the beam axis. The two runs differed only in the absorber material, defined through the Geant4 NIST database as `G4_Al` and `G4_Pb`; switching material required only recompiling the detector class.

The primary beam was defined through the General Particle Source as a monoenergetic point source of 5 GeV electrons placed in front of the block and directed along its longitudinal axis. Each run consisted of 1000 primary events, sufficient to suppress statistical fluctuations in the profiles while keeping the runtime short.

We also assessed the sensitivity of the results to the choice of electromagnetic physics list. In addition to the default `FTFP_BERT_EMZ` and `FTFP_BERT`, the simulation was repeated with the dedicated electromagnetic lists available in Geant4, namely the standard option (`G4EmStandardPhysics`) and its more refined variant (`G4EmStandardPhysics_option4`), which incorporates improved models for multiple scattering and low-energy processes. The longitudinal profiles and the extracted shower maxima were found to be consistent across all lists, with differences well within the statistical fluctuations of the 1000-event samples. This stability indicates that, for the energy and observables considered here, the gross features of the electromagnetic cascade are robust against the specific implementation of the underlying physics processes.

Energy deposition and particle production were recorded step by step through a custom `SteppingAction`. For each step the longitudinal coordinate along the beam axis, the energy deposited, and a flag identifying the first step of every new track were written to standard output

and collected into a per-material data file. The longitudinal coordinate was then shifted so that the front face of the block corresponds to the origin, converted to radiation-length units, and binned in slices of $0.1 X_0$. The energy profile dE/dt was obtained by weighting each entry by its deposited energy, while the multiplicity profile dN/dt was obtained by counting the first steps of new tracks.

Code Modifications

The **AnaEx01** example required three targeted modifications to adapt it to this measurement.

Detector geometry and material. The calorimeter parameters were hardcoded directly in `DetectorConstruction.cc`, immediately before the geometry is built, overriding the default macro-driven values:

```
fAbsorberThickness = 2500.0; // 250 cm (beam axis)
fGapThickness      = 0.0;
fCalorSizeYZ       = 2500.0; // 250 cm (transverse)
fNbOfLayers        = 1;
ComputeCalorParameters();

G4NistManager* nist = G4NistManager::Instance();
fAbsorberMaterial = nist->FindOrBuildMaterial("G4_Pb");
fGapMaterial      = nist->FindOrBuildMaterial("G4_Pb");
```

This produces a single homogeneous block of $250 \times 250 \times 250 \text{ cm}^3$. All dimensions are in millimetres (Geant4 internal units), so 2500.0 corresponds to 250 cm. Switching between aluminium and lead required changing the two material strings to `"G4_Al"` and recompiling, which took only a few seconds since a single translation unit was affected.

Longitudinal coordinate extraction. The standard `SteppingAction` was extended to extract the position of each step along the beam axis and log it together with the deposited energy and a new-track flag:

```
G4double z_depth =
aStep->GetPreStepPoint()->GetPosition().x();
G4int isNew =
```

```

(aStep->GetTrack()->GetCurrentStepNumber()==1)?1:0;
if (edep > 0. || isNew == 1) {
    G4cout << "DATA," << z_depth << ", "
    << edep    << ", "
    << isNew    << G4endl;
}

```

The x -coordinate corresponds to the longitudinal depth because the beam travels along the $+x$ axis. Printing to standard output allowed the data to be extracted without modifying the ROOT analysis chain, using a simple shell pipeline:

```

./AnaEx01 run.mac | grep "DATA," \
| sed 's/.*DATA/DATA/' > Lead_Shower.csv

```

Beam configuration. The primary beam was defined through the General Particle Source via the macro file `run.mac`:

```

/run/initialize
/gps/particle e-
/gps/energy 5 GeV
/gps/pos/type Point
/gps/pos/centre -26 0 0 cm
/gps/direction 1 0 0
/run/beamOn 1000

```

A monoenergetic 5 GeV electron was fired from a point source placed 26 cm upstream of the block front face along the beam axis, directed along $+x$. The source was placed outside the block to avoid initialising the primary track inside the absorbing material.

IV. RESULTS AND DISCUSSION

The reconstructed longitudinal profiles of deposited energy and secondary-particle multiplicity are shown in Fig. 1 as a function of depth $t = x/X_0$, with both distributions normalised to unit area.

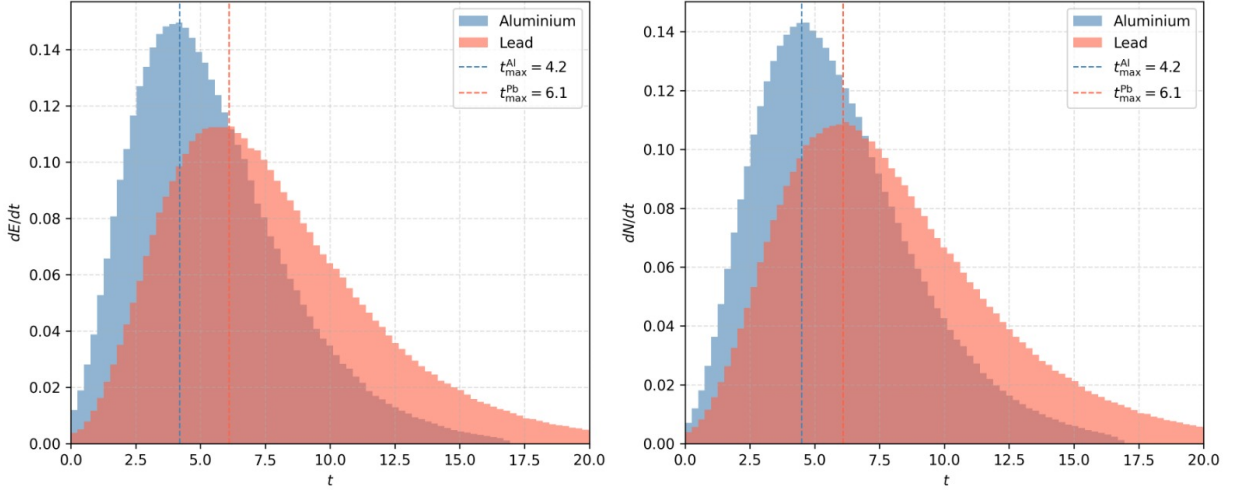


FIG. 1. Normalised longitudinal profiles of (a) deposited energy and (b) secondary-particle multiplicity for 5 GeV electrons in aluminium (solid) and lead (dashed) as a function of depth in radiation-length units $t = x/X_0$. Dotted vertical lines mark the analytical shower maxima of Table I.

Material	$t_{\max}^{\text{pred}}$	$t_{\max}^{\text{sim}}$
Aluminium	4.26	≈ 4.2
Lead	6.03	≈ 6.1

TABLE II. Predicted [Eq. (5)] and simulated shower maxima for 5 GeV electrons.

The shower maxima extracted from the simulation are compared with the analytical predictions in Table II. The agreement is good, with both materials reproduced to within $0.3 X_0$.

The residual offset between the simple model and the simulation is expected. The Rossi picture assumes an idealised binary splitting at fixed intervals and neglects the continuous ionization losses suffered by charged particles between radiative interactions; it therefore tends to overestimate the depth of the shower. The empirical formula of Eq. (5) partially corrects for this through the $C_{e\gamma}$ term, while the full Monte Carlo, which treats the realistic cross sections and continuous energy loss, places the maximum marginally closer to the front face for Aluminium and further from the front face for Lead.

Two features of Fig. 1 confirm the underlying physics. First, the energy and multiplicity profiles peak at the same depth in both materials, a direct consequence of the multiplicative nature of the cascade: the deposited energy is largest precisely where the number of secondary particles is greatest. Second, when expressed in X_0 units the two profiles are similar in shape, illustrating the approximate universality of the longitudinal development. The lead profile is nonetheless

broadener and exhibits a longer tail, consistent with its lower critical energy ($E_c^{\text{Pb}} \simeq 7.3$ MeV against $E_c^{\text{Al}} \simeq 42.7$ MeV), which sustains the multiplication over a greater number of radiation lengths before absorption dominates.

V. CONCLUSIONS

We have simulated the longitudinal development of 5 GeV electromagnetic showers in aluminium and lead with Geant4 and compared the results to analytical expectations. The simulated shower maxima agree with the predictions of the Rossi model and its empirical refinements to within $0.3 X_0$, the small residual offset being attributable to the continuous ionization losses neglected by the simple model. Expressed in radiation-length units, the longitudinal profiles of the two absorbers are approximately universal, and the remaining material dependence, a $\sim 1.8 X_0$ shift of the shower maximum, is fully accounted for by the difference in critical energy between aluminium and lead. These results validate the simulation setup and provide a quantitative benchmark for the use of Geant4 in electromagnetic calorimetry studies.

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- [1] B. Rossi and K. Greisen, *Rev. Mod. Phys.* **13**, 240 (1941).
 - [2] R. L. Workman *et al.* (Particle Data Group), *Prog. Theor. Exp. Phys.* **2022**, 083C01 (2022).
 - [3] S. Agostinelli *et al.* (Geant4 Collaboration), *Nucl. Instrum. Methods A* **506**, 250 (2003).